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Spring-REST

Create a Spring Web Project using Maven

Objective

Create a Spring Boot web application for REST services using Spring Web Framework.

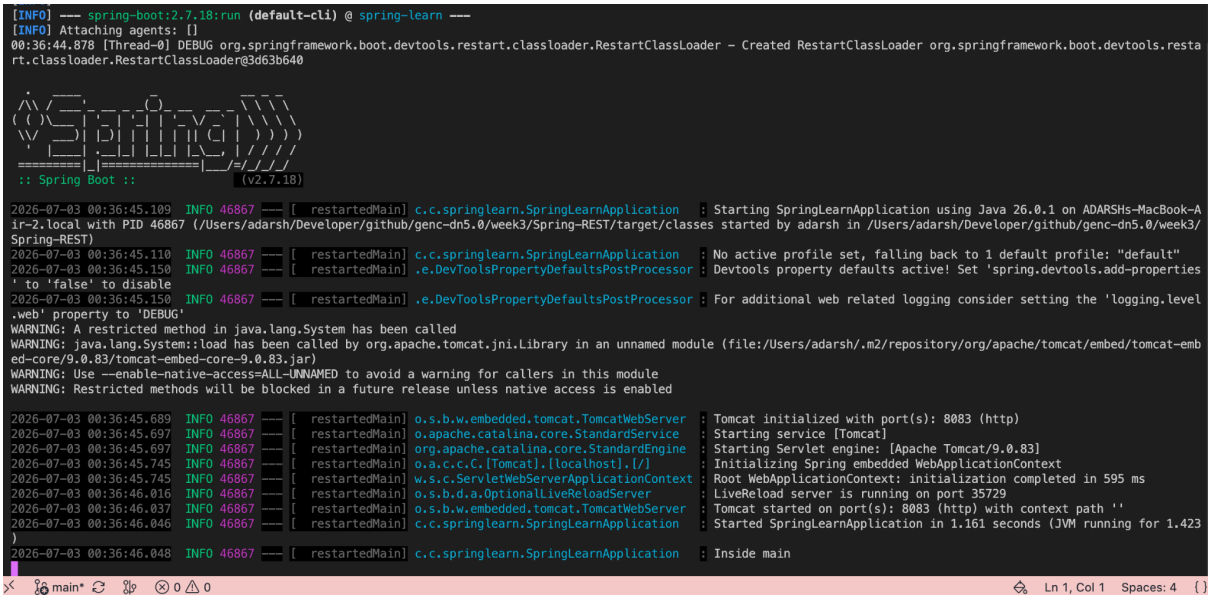
Steps

1. A Spring Boot project is created using start.spring.io with the following configurations:
- Group: com.cognizant
- Artifact: spring-learn
- Description: "Demo project for Spring REST"
- Dependencies: Spring Boot DevTools, Spring Web
2. The application.properties file is configured with the server port:
server.port=8083
3. The project is built using Maven and verified that the Spring Boot application starts successfully on port 8083.

Project Structure

```
spring-learn/
├── pom.xml
└── src/main/
    ├── java/com/cognizant/springlearn/
    │   ├── SpringLearnApplication.java
    │   ├── Country.java
    │   └── controller/
    │       ├── HelloController.java
    │       └── CountryController.java
    └── resources/
        ├── application.properties
        └── country.xml
```

Output:


[INFO] --- spring-boot:2.7.18:run (default-cli) @ spring-learn ---
[INFO] Attaching agents: []
00:36:44.878 [Thread-0] DEBUG org.springframework.boot.devtools.restart.classloader.RestartClassLoader - Created RestartClassLoader org.springframework.boot.devtools.restart.classloader.RestartClassLoader@3d63b640
:: Spring Boot :: (v2.7.18)
2026-07-03 00:36:45.109 INFO 46867 --- [restartedMain] c.c.springlearn.SpringLearnApplication : Starting SpringLearnApplication using Java 26.0.1 on ADARSHs-MacBook-Air-2.local with PID 46867 (/Users/adarsh/Developer/github/genc-dn5.0/week3/Spring-REST/target/classes started by adarsh in /Users/adarsh/Developer/github/genc-dn5.0/week3/Spring-REST)
2026-07-03 00:36:45.110 INFO 46867 --- [restartedMain] c.c.springlearn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-03 00:36:45.150 INFO 46867 --- [restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-03 00:36:45.150 INFO 46867 --- [restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by org.apache.tomcat.jni.Library in an unnamed module (file:/Users/adarsh/.m2/repository/org/apache/tomcat/embed/tomcat-embed-core/9.0.83/tomcat-embed-core-9.0.83.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled
2026-07-03 00:36:45.689 INFO 46867 --- [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port(s): 8083 (http)
2026-07-03 00:36:45.697 INFO 46867 --- [restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-03 00:36:45.697 INFO 46867 --- [restartedMain] org.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.83]
2026-07-03 00:36:45.745 INFO 46867 --- [restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-03 00:36:45.745 INFO 46867 --- [restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 595 ms
2026-07-03 00:36:46.016 INFO 46867 --- [restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-07-03 00:36:46.037 INFO 46867 --- [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port(s): 8083 (http) with context path ''
2026-07-03 00:36:46.046 INFO 46867 --- [restartedMain] c.c.springlearn.SpringLearnApplication : Started SpringLearnApplication in 1.161 seconds (JVM running for 1.423)
2026-07-03 00:36:46.048 INFO 46867 --- [restartedMain] c.c.springlearn.SpringLearnApplication : Inside main
main*
Ln 1, Col 1
Spaces: 4

Exercise 2: Spring Core – Load Country from Spring Configuration XML

Objective

Load country data from Spring XML configuration file and display the details.

Steps

1. A Spring XML configuration file named country.xml is created in src/main/resources directory.
2. A bean tag is defined in the Spring configuration for country with property and values:

```
<bean id="country" class="com.cognizant.springlearn.Country">
   <property name="code" value="IN"/>
   <property name="name" value="India"/>
</bean>
```

3. A list of all countries is defined using the <list> tag:

```
<bean id="countryList" class="java.util.ArrayList">
   <constructor-arg>
      <list>
         <ref bean="country"/>
         <bean class="com.cognizant.springlearn.Country">
            <property name="code" value="US"/>
            <property name="name" value="United States"/>
         </bean>
         <bean class="com.cognizant.springlearn.Country">
            <property name="code" value="JP"/>
            <property name="name" value="Japan"/>
         </bean>
         <bean class="com.cognizant.springlearn.Country">
            <property name="code" value="DE"/>
            <property name="name" value="Germany"/>
         </bean>
      </list>
   </constructor-arg>
</bean>
```

4. A Country class is created with:
- Instance variables for code and name

- Empty parameter constructor with debug log message "Inside Country Constructor."
- Getters and setters with debug logs
- toString() method

5. A displayCountry() method is added in SpringLearnApplication.java that reads the country bean from Spring configuration file using ClassPathXmlApplicationContext and displays the country details.
6. The displayCountry() method is invoked in the main() method of SpringLearnApplication.java.
7. The application is executed and the logs are checked to verify which constructors and methods were invoked.

Hello World RESTful Web Service

Objective

Create a REST service that returns the text "Hello World!!" using Spring Web Framework.

Steps

1. A new package com.cognizant.springlearn.controller is created.
2. A class HelloController is created with @RestController annotation.
3. A method sayHello() is added with @GetMapping("/hello") annotation that returns the hardcoded string "Hello World!!".
4. Start and end logs are included in the sayHello() method.
5. The application is executed and the URL http://localhost:8083/hello is tested in both Chrome browser and Postman.

Controller Code

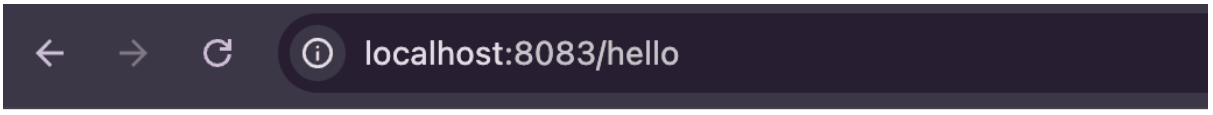
```
@RestController
public class HelloController {

    private static final Logger LOGGER = LoggerFactory.getLogger(HelloController.class);

    @GetMapping("/hello")
    public String sayHello() {
        LOGGER.info("Start");
        LOGGER.info("End");
        return "Hello World!!";
    }
}
```

Sample Request

```
GET http://localhost:8083/hello
```

Sample Response

```
Hello World!!
```



Hello World!!

REST - Country Web Service

Objective

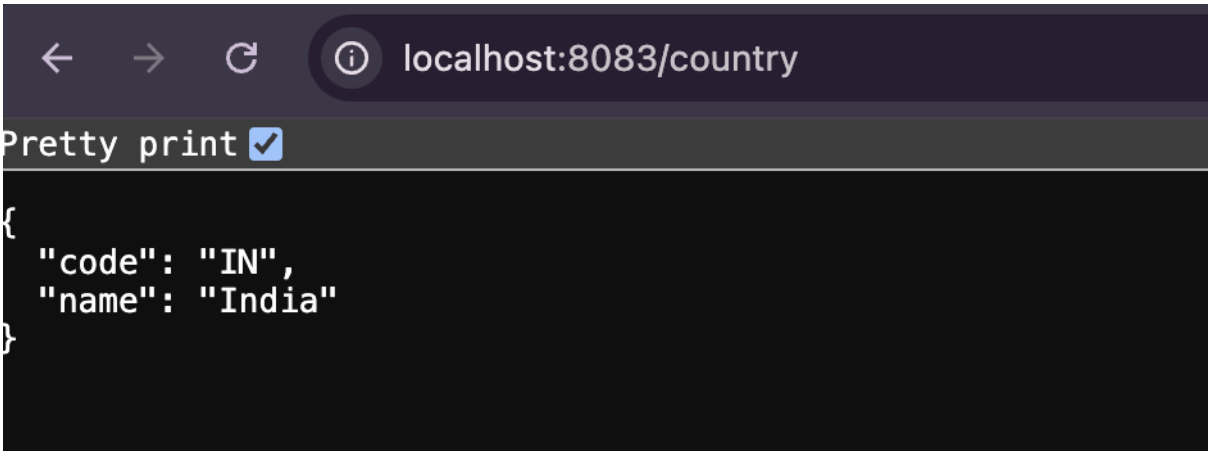
Create a REST service that returns India country details.

Steps

1. A new class CountryController is created in com.cognizant.springlearn.controller package with @RestController annotation.
2. A method getCountryIndia() is added with @RequestMapping("/country") annotation that loads the India bean from country.xml and returns it.
3. The application is executed and the URL http://localhost:8083/country is tested.

Controller Code

```
@RestController
public class CountryController {

    private static final Logger LOGGER = LoggerFactory.getLogger(CountryController.class);

    @RequestMapping("/country")
    public Country getCountryIndia() {
        LOGGER.info("Start");
        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
        Country country = (Country) context.getBean("country", Country.class);
        LOGGER.info("End");
        return country;
    }
}
```



REST - Get All Countries

Objective

Create a REST service that returns all the countries.

Steps

1. A method getAllCountries() is added in CountryController with @GetMapping("/countries") annotation that loads the country list from country.xml and returns it.
2. The application is executed and the URL http://localhost:8083/countries is tested.

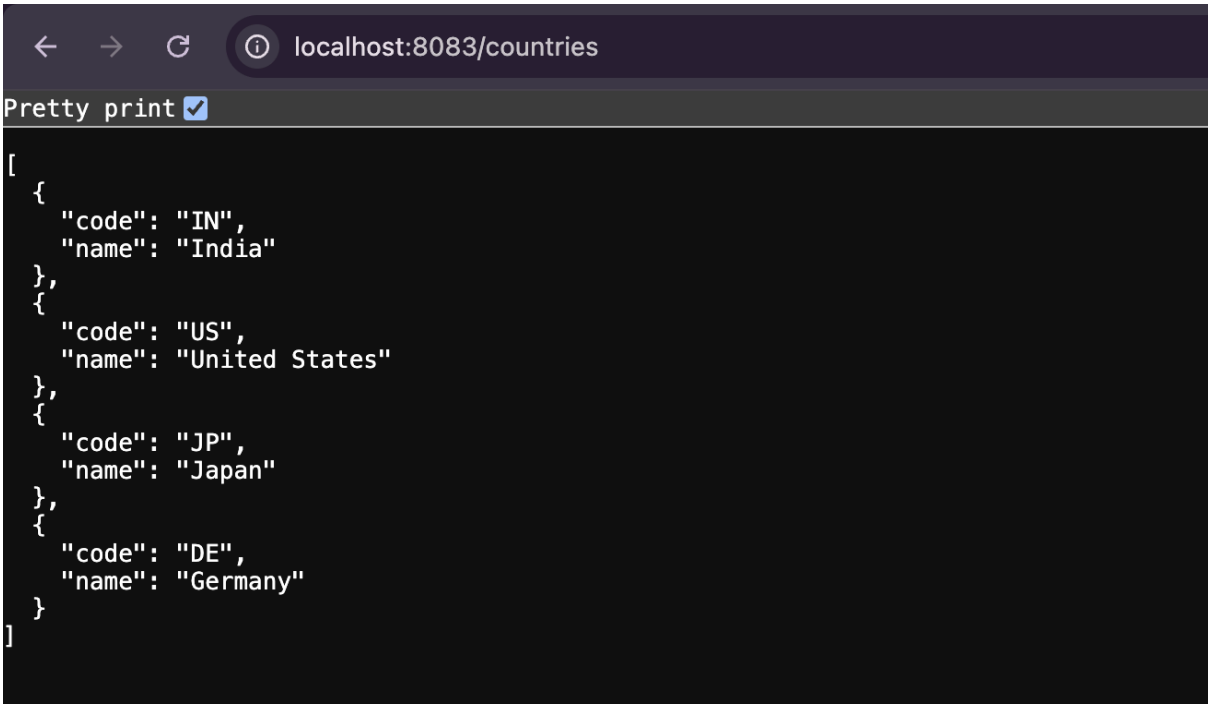
Controller Code

```
@GetMapping("/countries")
public List<Country> getAllCountries() {
   LOGGER.info("Start");
   ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
   @SuppressWarnings("unchecked")
   List<Country> countries = (List<Country>) context.getBean("countryList");
   LOGGER.info("End");
   return countries;
}
```

Sample Request

GET http://localhost:8083/countries

Sample Response



REST - Get Country by Code

Objective

Create a REST service that returns a country based on its two-character ISO code.

Steps

1. A method getCountryByCode() is added in CountryController with @GetMapping("/countries/{code}") annotation that loads the country list from country.xml, filters by code, and returns the matching country.
2. The @PathVariable annotation is used to extract the code from the URL path.
3. The application is executed and URLs like http://localhost:8083/countries/IN and http://localhost:8083/countries/JP are tested.

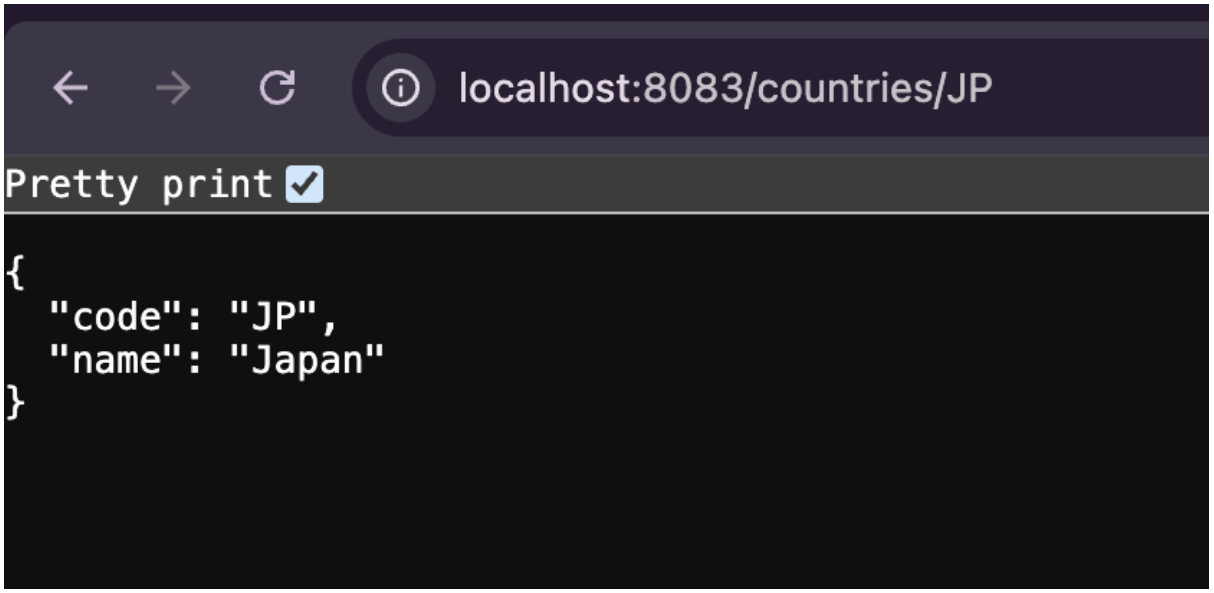
Controller Code

```
@GetMapping("/countries/{code}")
public Country getCountryByCode(@PathVariable String code) {
    LOGGER.info("Start");
    ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
    @SuppressWarnings("unchecked")
    List<Country> countries = (List<Country>) context.getBean("countryList");
    Country result = countries.stream()
            .filter(c -> c.getCode().equalsIgnoreCase(code))
            .findFirst()
            .orElse(null);
    LOGGER.debug("Country found for code {}: {}", code, result);
    LOGGER.info("End");
    return result;
}
```

Sample Requests

```
GET http://localhost:8083/countries/IN
GET http://localhost:8083/countries/US
GET http://localhost:8083/countries/JP
GET http://localhost:8083/countries/DE
```

Sample Response


localhost:8083/countries/JP
Pretty print
{
"code": "JP",
"name": "Japan"
}